HS Data Science 26

Session 01: Orientation, Topics, and Workflow

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Seminar Data Science

Focus of This Year's Seminar

- retrieval-augmented generation
- agentic search
- open web retrieval on our own infrastructure

Core Idea

This seminar rewards students who can formulate an interesting question, design a manageable study, and defend their choices clearly.

Speaker notes

Open with the change in seminar logic: fewer predefined topics, stronger emphasis on originality, question quality, and communication.

Seminar Topic

The seminar focuses on retrieval-augmented generation and smart agents in the context of open web search.

Technical Context

- we run our own search engine on the Open Web Index
- students can explore topics through this infrastructure
- the environment supports retrieval experiments and agent workflows

Research Perspective

- search is part of the research object
- retrieval choices can be inspected and compared
- the seminar goes beyond a generic RAG demo

Speaker notes

Position the seminar technically: shift from generic ML topics to a coherent research environment built on open retrieval and agentic workflows.

Why OWI and Our Search Engine?

The OWI-based environment exists because web retrieval research is otherwise structurally constrained by proprietary search engines and black-box APIs.

What This Enables

- inspect retrieval behaviour
- compare sparse, dense, and hybrid retrieval
- study provenance and ranking choices
- build agentic workflows on transparent search

Consequence for the Seminar

- the search engine is not just a utility
- the retrieval setup itself can be analysed
- good topics exploit the openness of the system

Demo Search

Use the OURRS demo web search for demonstrations and experiments.

Data and Statistics

- Open Web Index

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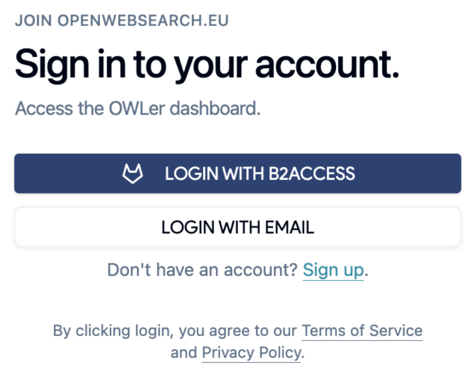
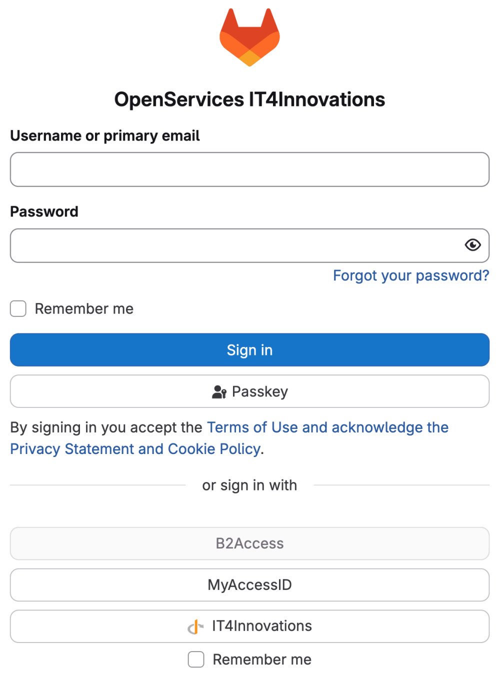
API & Data Download

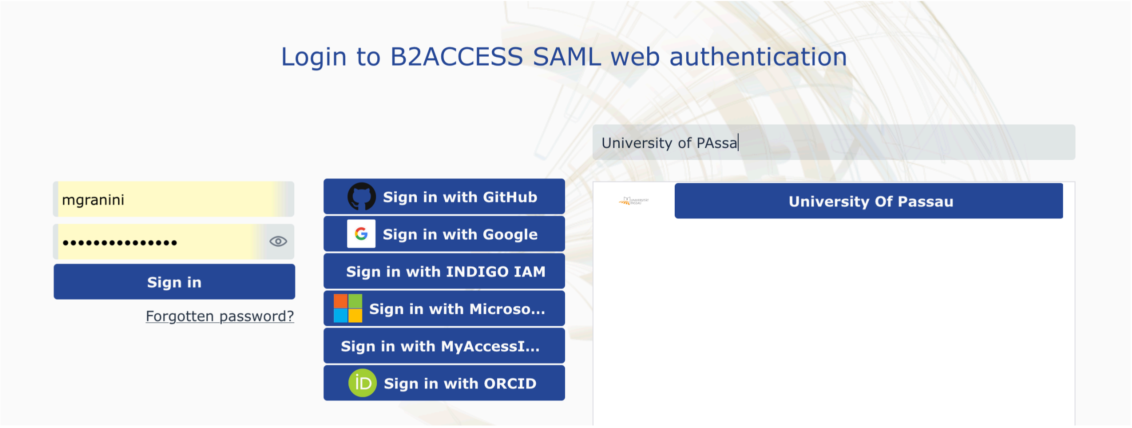
Direct dataL Corpora that could be used: [Open Web Search Curlie 2025]

API Access - Developer Guide:

- https://ourrs.eu/developers
- Login into dashboard: use B2ACCESS and your University of Passau Credentials







How Topic Selection Works

You are expected to generate your own research direction within the seminar theme.

Good Starting Angles

- retrieval quality
- agent planning and verification
- trust and usability
- corpus and query analysis

What Makes a Good Topic

- interesting research question
- manageable scope
- clear evaluation strategy
- relevance to the seminar setup

Example

- weak: *"Build a RAG demo"*
- better: *"When does query reformulation improve agentic search on exploratory tasks?"*

Expected Seminar Work

Scientific Working

- conduct paper research
- identify the gap or tension
- formulate research questions
- document assumptions and limits

Methodology

- implement a prototype or workflow
- run small-scale experiments
- compare alternatives where useful
- explain why the evidence supports the claim

> A modest experiment with a sharp question and clean reasoning is better than an oversized prototype without a defensible result.

Grading

Component	Weight
Independent idea generation	20%
Small-scale but rigorous experiments	15%
Clear methodology and evaluation	15%
Presentation	40%
Report	10%

The asymmetry is intentional: idea generation and presentation are the parts students cannot outsource blindly to AI.

Deliverables

Expected Outputs

- a 1-pager research narrative
- a seminar presentation with a clear research narrative
- a concise report in ACM format
- a Git repository with code, prompts, experiments, or analysis artefacts

Practical Rule

Keep every artefact aligned with one central claim: what question you asked, how you investigated it, and what you learned.

AI Usage

- **Opt-In**: you can use AI for everything, but you have to make it transparent.
- If you use AI, expectations on results, code, report and presentation will be **significantly higher**
- **You fail**
  - if you use AI, but do not make it transparent.
  - if your AI did something wrong (even only a tiny thing) and you did not recognize it.
    - A wrong citation
    - a wrong experiment
    - a wrong claim
    - etc.
  - If your AI did something, and you have
    - no idea what
    - cannot explain concepts, methods, results
    - cannot explain why it did what it did

Organization

Study Programmes

- BSc Computer Science
- BSc Internet Computing
- MSc Computer Science
- MSc AI Engineering
- open to other faculties

Semester Rhythm

- orientation and topic exploration
- question development
- prototype implementation
- presentations with feedback
- refinement and final report

> Lock the question early, then iterate on evidence and presentation quality.

Tentative Schedule

Date	Topic
Tue 14/04/26	Topic Presentation, Research Hypothesis
Tue 21/04/26	Scientific Working I
Tue 28/04/26	No Seminar
Tue 05/05/26	Scientific Working II
Tue 12/05/26	Research Hypothesis Presentation
Tue 19/05 – 23/06	Q&A sessions
Tue 30/06 – 14/07	Student Presentations

All sessions: 10:00–12:00, ITZ R 138.

Next Steps

Before the first week of semester, students should:

- research the field of RAG, agentic search, and web search
- write down one page with a possible topic or research direction
- submit it or present it in the first week of semester

Research-hypothesis presentations are 5–10 minutes.

Spots will be assigned based on the quality of the proposal.

Looking Ahead

Before the next milestone, each student should prepare:

- three candidate topic ideas
- one tentative main research question
- one plausible evaluation strategy
- one short argument for why the question is interesting

> This seminar is about producing an interesting, defensible piece of research in an AI-rich environment where originality, judgment, and communication matter more than routine implementation.